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Real-Time Training-Free Multimodal Identity Verification using Biometrical Score Fusion

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The Problem

Unimodal Biometrics Can Fail in the Real World

Face recognition

- Sensitive to illumination changes
- Degrades under pose variation
- Fails on motion blur & occlusion

Voice recognition

- Degrades under ambient noise
- Struggles with reverberant rooms
- Vulnerable to cross-speaker confusion

Results in elevated false rejection and false acceptance rates in deployment

Motivation

The Gap in Existing Multimodal Fusion

Paired Dataset → **Joint Training** → **Fused Model**

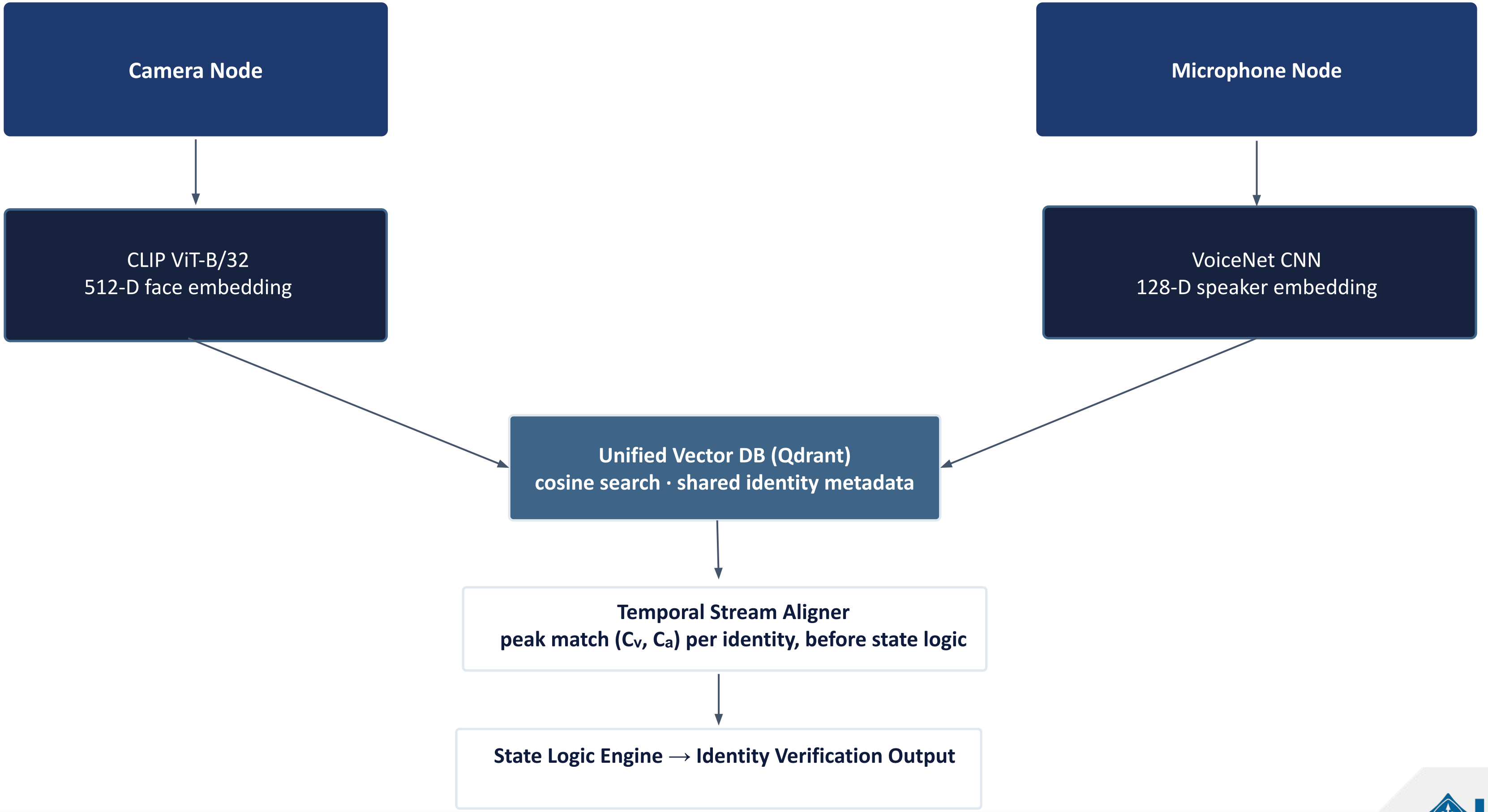
- Deep multimodal fusion needs synchronized, paired datasets
- Joint training adds computational overhead and overfitting risk
- Prevents independent upgrades of face / voice sub-models

Real-Time Training-Free Multimodal Identity Verification using Biometrical Score Fusion

Contributions

1. Real-time multimodal framework fusing face and voice biometrics
2. Unified Qdrant retrieval backend for heterogeneous embeddings
3. Training-free fusion - no joint optimization or paired data required
4. Temporal decision layer: agreement, missing-stream, and conflict states

SYSTEM DESIGN



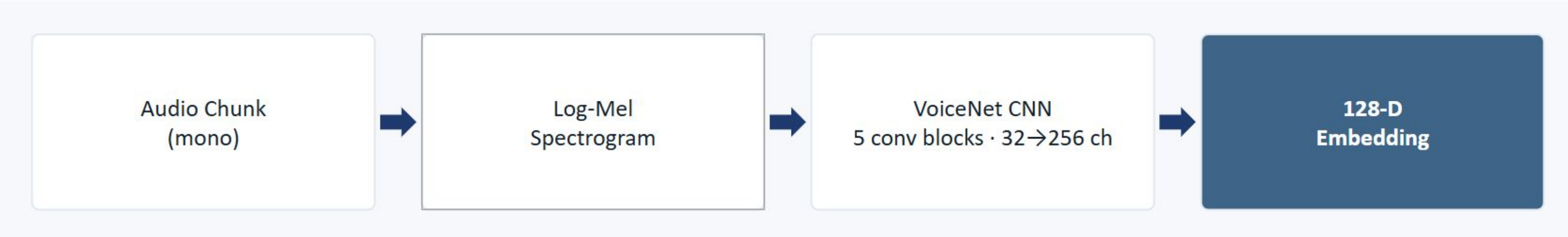
Visual pipeline

Face recognition algorithm

- Builds on prior zero-shot work (Labinjan et al.) - encoder unchanged
- Robust to domain shifts: pose, illumination and blur variation
- Migrated retrieval backend from FAISS to Qdrant for unified infrastructure

Audio pipeline

Voice recognition algorithm



- Trained closed-set on VoxCeleb: Conv2d + BatchNorm + MaxPool x5
- The “truncation trick”: Softmax head used only for training, then discarded
- Remaining backbone reused as zero-shot open-set speaker embedding extractor

Infrastructure

Qdrant unification

Two Separate FAISS Indexes	One Unified Qdrant Backend
Two vector indexes to maintain	Centralized vector storage and retrieval
Duplicated identity metadata	Shared identity and metadata schema
Manual sync across storage layers	Modality-specific similarity retained
Separate persistence mechanisms	One deployment configuration

Clarification: Qdrant unifies infrastructure and identity metadata - not the vector spaces themselves. Face (512-D) and voice (128-D) embeddings remain separate.

Methodology

Score alignment and normalization

Camera and microphone streams arrive asynchronously at different rates - strict frame-level alignment is infeasible. Instead, the system takes the peak similarity score per modality within an evaluation window.

V

Visual peak score

$$S_V = \max\{ s_{v1}, s_{v2}, \ldots, s_{vn} \}$$

A

Auditory peak score

$$S_A = \max\{ s_{a1}, s_{a2}, \ldots, s_{am} \}$$

Eq. (1)–(2): both pathways share Qdrant cosine-similarity retrieval, so S_v and S_a are compatible and can be thresholded before fusion.

Experimental setup

Dataset & evaluation protocol

```
known_data/
    identity_i/
        train/  (enrollment gallery)
            audio/
            vision/
        val/  (evaluation probes)
            audio/
            vision/
```

Three probe outcomes

Correct identification

prediction matches ground truth

Misidentification

system accepts a wrong identity

Rejection

outputs UNKNOWN or CONFLICT

Methodology

Evaluation metrics

Accuracy	Misidentification Rate	Rejection Rate (unknown)
n_correct/n_total	n_misidentified/n_total	n_rejected/n_total

Together, the three quantities describe complementary failure behavior across the unimodal and fused configurations.

Results

Comparison of unimodal and fused biometric verification configurations

Configuration	Accuracy	Misidentification rate	Rejection rate
Face-only	92.00%	0.00%	8.00%
Voice-only	83.00%	9.00%	7.00%
Fused	75.00%	0.00%	25.00%

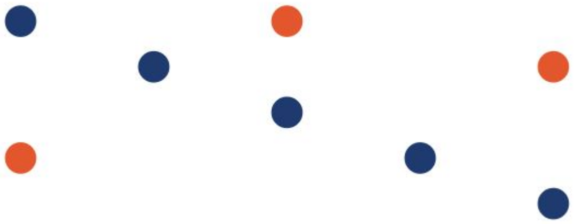
Analysis

Complementary error patterns

Face-only errors: sparse

Errors mostly appear as rejections under pose, scale, or quality variation - rarely cross-identity.

Voice-only errors: scattered



Errors scatter across multiple enrolled identities - the acoustic subsystem sometimes confuses speakers.

Non-overlapping failure modes: when audio misidentifies, vision usually gives the correct match or UNKNOWN - so the CONFLICT state catches the contradiction.

Correct-identity error (rejection)

Cross-identity misidentification

Conclusion & future work

Contributions

What We Did

- ✓ Training-free multimodal fusion - no paired audio-visual data, no fusion network to train
- ✓ Unified Qdrant retrieval backend - merges 512-D CLIP (face) and 128-D VoiceNet (voice) embeddings under one identity-aware infrastructure, without forcing them into a shared latent space
- ✓ Deterministic state-based fusion layer - explicit CONFIRMED / VISION_ONLY / AUDIO_ONLY / CONFLICT logic for agreement, missing streams, and cross-modal conflicts
- ✓ Misidentification rate cut from 9.00% → 0.00% by rejecting under cross-modal contradiction - a deliberate, risk-averse security trade-off (accuracy 83% → 75%, rejection 7% → 25%)

Future Steps

- Edge-device deployment
- Latency and resource profiling
- Live cross-modal synchronization